

VESSEL ANALYSIS REPORT

vaf(%)	branch points count	tip point count	vein count	total vein length	average vein length
2.215	102.0	183.0	244.0	6391.67	26.195

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
1	[228,0]	[235,8]	9.899	TIP	BRANCH
2	[235,8]	[241,8]	5.0	BRANCH	BRANCH
3	[235,8]	[212,12]	24.071	BRANCH	BRANCH
4	[249,2]	[241,8]	9.071	TIP	BRANCH
5	[257,51]	[241,8]	48.213	TIP	BRANCH
6	[30,10]	[8,32]	41.243	BRANCH	BRANCH
7	[30,10]	[92,49]	85.77	BRANCH	BRANCH
8	[35,16]	[30,10]	6.657	TIP	BRANCH
9	[142,11]	[153,11]	10.0	BRANCH	BRANCH
10	[142,11]	[92,49]	92.184	BRANCH	BRANCH
11	[138,19]	[142,11]	8.243	TIP	BRANCH
12	[190,12]	[153,11]	43.627	TIP	BRANCH
13	[163,3]	[153,11]	11.899	TIP	BRANCH
14	[204,4]	[212,12]	12.071	TIP	BRANCH
15	[212,12]	[105,73]	157.154	BRANCH	BRANCH
16	[356,0]	[351,17]	18.071	TIP	BRANCH
17	[349,18]	[351,17]	1.414	TIP	BRANCH
18	[353,18]	[351,17]	1.0	TIP	BRANCH
19	[434,1]	[446,24]	27.556	TIP	BRANCH
20	[458,31]	[446,24]	13.899	TIP	BRANCH
21	[446,24]	[424,34]	24.728	BRANCH	BRANCH
22	[505,17]	[489,29]	22.899	TIP	BRANCH
23	[477,21]	[489,29]	14.314	TIP	BRANCH
24	[489,29]	[477,63]	42.799	BRANCH	BRANCH
25	[1,40]	[8,32]	10.485	TIP	BRANCH
26	[8,32]	[19,53]	24.728	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
27	[365,21]	[424,34]	63.385	TIP	BRANCH
28	[424,34]	[423,49]	14.414	BRANCH	BRANCH
29	[92,49]	[94,65]	15.828	BRANCH	BRANCH
30	[421,50]	[423,49]	1.0	TIP	BRANCH
31	[425,51]	[423,49]	1.414	TIP	BRANCH
32	[19,53]	[47,95]	56.698	BRANCH	BRANCH
33	[7,54]	[19,53]	11.414	TIP	BRANCH
34	[342,83]	[397,57]	67.698	TIP	BRANCH
35	[397,57]	[454,77]	85.113	BRANCH	BRANCH
36	[400,55]	[397,57]	2.414	TIP	BRANCH
37	[467,52]	[477,63]	14.728	TIP	BRANCH
38	[477,63]	[476,77]	13.414	BRANCH	BRANCH
39	[94,65]	[105,73]	13.899	BRANCH	BRANCH
40	[94,65]	[55,95]	53.941	BRANCH	BRANCH
41	[111,99]	[105,73]	27.485	TIP	BRANCH
42	[449,69]	[454,77]	9.071	TIP	BRANCH
43	[454,77]	[476,77]	21.0	BRANCH	BRANCH
44	[476,77]	[489,106]	33.971	BRANCH	BRANCH
45	[47,95]	[55,95]	7.0	BRANCH	BRANCH
46	[42,98]	[47,95]	4.828	TIP	BRANCH
47	[55,95]	[56,98]	2.414	BRANCH	BRANCH
48	[56,98]	[84,143]	59.698	BRANCH	BRANCH
49	[56,98]	[63,143]	48.556	BRANCH	BRANCH
50	[489,106]	[489,109]	2.0	BRANCH	BRANCH
51	[510,74]	[489,106]	42.213	TIP	BRANCH
52	[489,109]	[488,146]	36.0	BRANCH	BRANCH
53	[504,118]	[489,109]	17.314	TIP	BRANCH
54	[5,110]	[21,121]	20.728	TIP	BRANCH
55	[21,121]	[63,143]	50.113	BRANCH	BRANCH
56	[21,121]	[36,162]	46.213	BRANCH	BRANCH
57	[433,130]	[432,137]	6.414	TIP	BRANCH
58	[432,137]	[470,151]	42.799	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
59	[432,137]	[414,144]	19.899	BRANCH	BRANCH
60	[63,143]	[64,151]	7.414	BRANCH	BRANCH
61	[89,144]	[84,143]	4.414	TIP	BRANCH
62	[84,143]	[64,151]	21.899	BRANCH	BRANCH
63	[408,142]	[414,144]	5.828	TIP	BRANCH
64	[414,144]	[388,174]	46.213	BRANCH	BRANCH
65	[499,146]	[488,146]	10.828	TIP	BRANCH
66	[488,146]	[470,151]	18.657	BRANCH	BRANCH
67	[64,151]	[58,164]	14.485	BRANCH	BRANCH
68	[470,151]	[471,179]	27.414	BRANCH	BRANCH
69	[470,151]	[422,180]	60.941	BRANCH	BRANCH
70	[161,149]	[159,152]	2.828	TIP	BRANCH
71	[139,139]	[159,152]	24.971	TIP	BRANCH
72	[170,161]	[159,152]	13.899	TIP	BRANCH
73	[36,162]	[58,164]	21.828	BRANCH	BRANCH
74	[36,162]	[15,189]	34.87	BRANCH	BRANCH
75	[69,216]	[58,164]	56.385	TIP	BRANCH
76	[388,174]	[374,190]	23.142	BRANCH	BRANCH
77	[382,168]	[388,174]	7.071	TIP	BRANCH
78	[471,179]	[422,180]	55.042	BRANCH	BRANCH
79	[471,179]	[499,218]	57.971	BRANCH	BRANCH
80	[418,181]	[422,180]	3.414	TIP	BRANCH
81	[1,186]	[15,189]	15.071	TIP	BRANCH
82	[15,189]	[18,218]	29.243	BRANCH	BRANCH
83	[375,98]	[374,190]	127.711	TIP	BRANCH
84	[374,190]	[372,208]	17.414	BRANCH	BRANCH
85	[364,211]	[372,208]	7.828	TIP	BRANCH
86	[372,208]	[378,219]	12.071	BRANCH	BRANCH
87	[24,222]	[18,218]	6.657	TIP	BRANCH
88	[0,210]	[18,218]	20.899	TIP	BRANCH
89	[406,218]	[409,218]	2.0	BRANCH	BRANCH
90	[406,218]	[383,220]	22.828	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
91	[403,230]	[406,218]	12.243	TIP	BRANCH
92	[429,201]	[409,218]	25.627	TIP	BRANCH
93	[409,218]	[428,224]	20.071	BRANCH	BRANCH
94	[499,218]	[463,225]	37.899	BRANCH	BRANCH
95	[500,221]	[499,218]	2.0	TIP	BRANCH
96	[378,219]	[383,220]	4.414	BRANCH	BRANCH
97	[265,229]	[378,219]	127.669	TIP	BRANCH
98	[367,260]	[383,220]	45.213	TIP	BRANCH
99	[428,224]	[456,225]	31.971	BRANCH	BRANCH
100	[428,224]	[430,236]	11.414	BRANCH	BRANCH
101	[456,225]	[463,225]	6.0	BRANCH	BRANCH
102	[456,225]	[430,236]	29.142	BRANCH	BRANCH
103	[463,225]	[458,260]	36.899	BRANCH	BRANCH
104	[430,236]	[428,247]	10.828	BRANCH	BRANCH
105	[103,237]	[104,240]	2.414	TIP	BRANCH
106	[169,213]	[104,240]	84.456	TIP	BRANCH
107	[104,240]	[62,260]	48.87	BRANCH	BRANCH
108	[428,247]	[410,253]	19.485	BRANCH	BRANCH
109	[428,247]	[458,260]	33.971	BRANCH	BRANCH
110	[410,253]	[359,284]	65.184	BRANCH	BRANCH
111	[410,253]	[415,277]	24.657	BRANCH	BRANCH
112	[62,260]	[41,276]	28.385	BRANCH	BRANCH
113	[56,244]	[62,260]	17.071	TIP	BRANCH
114	[458,260]	[458,273]	12.0	BRANCH	BRANCH
115	[458,273]	[464,280]	9.657	BRANCH	BRANCH
116	[458,273]	[423,285]	38.971	BRANCH	BRANCH
117	[32,256]	[37,275]	20.071	TIP	BRANCH
118	[37,275]	[41,276]	3.414	BRANCH	BRANCH
119	[37,275]	[19,278]	18.243	BRANCH	BRANCH
120	[104,283]	[41,276]	90.012	TIP	BRANCH
121	[415,277]	[423,285]	11.485	BRANCH	BRANCH
122	[400,292]	[415,277]	23.728	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
123	[1,263]	[19,278]	24.971	TIP	BRANCH
124	[20,293]	[19,278]	14.414	TIP	BRANCH
125	[464,280]	[499,287]	36.899	BRANCH	BRANCH
126	[464,280]	[439,318]	49.113	BRANCH	BRANCH
127	[359,284]	[342,311]	34.799	BRANCH	BRANCH
128	[305,261]	[359,284]	68.698	TIP	BRANCH
129	[423,285]	[394,326]	56.87	BRANCH	BRANCH
130	[499,287]	[494,345]	64.042	BRANCH	BRANCH
131	[509,255]	[499,287]	34.728	TIP	BRANCH
132	[342,311]	[342,341]	29.828	BRANCH	BRANCH
133	[318,306]	[342,311]	24.657	TIP	BRANCH
134	[439,318]	[421,333]	26.728	BRANCH	BRANCH
135	[448,332]	[439,318]	16.314	TIP	BRANCH
136	[394,326]	[375,347]	31.971	BRANCH	BRANCH
137	[394,326]	[414,336]	22.728	BRANCH	BRANCH
138	[421,333]	[414,336]	7.243	BRANCH	BRANCH
139	[421,333]	[442,351]	27.042	BRANCH	BRANCH
140	[414,336]	[414,341]	4.0	BRANCH	BRANCH
141	[199,317]	[160,339]	51.355	TIP	BRANCH
142	[150,340]	[160,339]	9.414	TIP	BRANCH
143	[162,342]	[160,339]	2.828	TIP	BRANCH
144	[342,341]	[375,347]	34.485	BRANCH	BRANCH
145	[342,341]	[338,349]	8.657	BRANCH	BRANCH
146	[414,341]	[375,351]	45.456	BRANCH	BRANCH
147	[414,341]	[441,355]	33.142	BRANCH	BRANCH
148	[510,365]	[494,345]	28.556	TIP	BRANCH
149	[494,345]	[446,372]	57.77	BRANCH	BRANCH
150	[375,347]	[375,351]	3.0	BRANCH	BRANCH
151	[37,347]	[37,349]	1.0	TIP	BRANCH
152	[37,349]	[44,351]	6.828	BRANCH	BRANCH
153	[3,358]	[37,349]	36.314	TIP	BRANCH
154	[338,349]	[292,354]	48.728	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
155	[338,359]	[338,349]	9.0	TIP	BRANCH
156	[74,328]	[44,351]	40.284	TIP	BRANCH
157	[44,351]	[44,356]	4.0	BRANCH	BRANCH
158	[375,351]	[369,372]	23.071	BRANCH	BRANCH
159	[442,351]	[441,355]	3.414	BRANCH	BRANCH
160	[472,326]	[442,351]	41.87	TIP	BRANCH
161	[292,354]	[302,385]	38.142	BRANCH	BRANCH
162	[288,348]	[292,354]	6.243	TIP	BRANCH
163	[441,355]	[446,372]	17.657	BRANCH	BRANCH
164	[75,393]	[44,356]	53.77	TIP	BRANCH
165	[40,361]	[44,356]	6.243	TIP	BRANCH
166	[369,372]	[392,391]	32.213	BRANCH	BRANCH
167	[369,372]	[363,391]	20.657	BRANCH	BRANCH
168	[446,372]	[405,392]	48.456	BRANCH	BRANCH
169	[15,399]	[22,382]	18.899	TIP	BRANCH
170	[8,380]	[22,382]	14.243	TIP	BRANCH
171	[28,379]	[22,382]	5.828	TIP	BRANCH
172	[275,412]	[302,385]	46.213	TIP	BRANCH
173	[302,385]	[319,404]	26.385	BRANCH	BRANCH
174	[363,391]	[392,391]	30.485	BRANCH	BRANCH
175	[363,391]	[342,408]	27.627	BRANCH	BRANCH
176	[392,391]	[405,392]	12.414	BRANCH	BRANCH
177	[405,392]	[410,418]	27.071	BRANCH	BRANCH
178	[465,387]	[466,394]	6.414	TIP	BRANCH
179	[460,401]	[466,394]	8.657	TIP	BRANCH
180	[490,377]	[466,394]	83.326	TIP	BRANCH
181	[325,405]	[319,404]	5.414	TIP	BRANCH
182	[338,407]	[342,408]	3.414	TIP	BRANCH
183	[342,408]	[346,428]	20.657	BRANCH	BRANCH
184	[410,418]	[423,418]	12.0	BRANCH	BRANCH
185	[410,418]	[397,425]	14.485	BRANCH	BRANCH
186	[372,497]	[423,418]	110.74	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
187	[429,402]	[423,418]	19.314	TIP	BRANCH
188	[397,425]	[346,428]	52.899	BRANCH	BRANCH
189	[383,452]	[397,425]	33.213	TIP	BRANCH
190	[346,428]	[344,432]	3.828	BRANCH	BRANCH
191	[312,431]	[344,432]	33.071	TIP	BRANCH
192	[339,485]	[344,432]	58.799	TIP	BRANCH
193	[291,436]	[278,438]	12.828	TIP	BRANCH
194	[278,445]	[278,438]	6.0	TIP	BRANCH
195	[230,387]	[278,438]	102.456	TIP	BRANCH
196	[506,448]	[467,484]	68.87	TIP	BRANCH
197	[478,483]	[467,484]	10.414	TIP	BRANCH
198	[467,484]	[461,486]	5.828	BRANCH	BRANCH
199	[462,492]	[461,486]	5.414	TIP	BRANCH
200	[452,482]	[461,486]	9.243	TIP	BRANCH
201	[462,0]	[472,0]	11.657	TIP	TIP
202	[485,0]	[492,6]	10.657	TIP	TIP
203	[285,31]	[284,50]	21.899	TIP	TIP
204	[258,64]	[254,71]	8.657	TIP	TIP
205	[301,90]	[273,92]	29.657	TIP	TIP
206	[210,102]	[211,104]	2.414	TIP	TIP
207	[388,107]	[391,108]	3.414	TIP	TIP
208	[225,108]	[237,110]	12.828	TIP	TIP
209	[182,118]	[177,120]	5.828	TIP	TIP
210	[253,126]	[289,128]	37.657	TIP	TIP
211	[374,133]	[364,143]	14.728	TIP	TIP
212	[268,150]	[245,160]	27.971	TIP	TIP
213	[100,155]	[92,212]	63.627	TIP	TIP
214	[143,160]	[133,175]	19.142	TIP	TIP
215	[284,163]	[279,171]	11.243	TIP	TIP
216	[228,217]	[233,221]	6.657	TIP	TIP
217	[214,232]	[208,239]	9.485	TIP	TIP
218	[149,266]	[141,272]	10.485	TIP	TIP

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
219	[196,277]	[194,286]	9.828	TIP	TIP
220	[229,279]	[249,286]	23.728	TIP	TIP
221	[288,287]	[283,290]	6.243	TIP	TIP
222	[119,290]	[114,307]	19.071	TIP	TIP
223	[19,303]	[2,314]	22.142	TIP	TIP
224	[216,317]	[208,318]	8.414	TIP	TIP
225	[306,321]	[253,346]	69.355	TIP	TIP
226	[219,343]	[217,349]	6.828	TIP	TIP
227	[127,349]	[108,350]	21.071	TIP	TIP
228	[96,381]	[111,389]	18.314	TIP	TIP
229	[140,406]	[137,413]	8.243	TIP	TIP
230	[69,415]	[62,421]	10.657	TIP	TIP
231	[445,419]	[452,424]	9.071	TIP	TIP
232	[32,432]	[31,439]	8.243	TIP	TIP
233	[48,457]	[46,463]	6.828	TIP	TIP
234	[109,460]	[117,462]	8.828	TIP	TIP
235	[0,462]	[16,509]	54.213	TIP	TIP
236	[450,464]	[437,466]	13.828	TIP	TIP
237	[272,465]	[270,472]	7.828	TIP	TIP
238	[362,466]	[359,469]	4.828	TIP	TIP
239	[511,472]	[467,510]	62.669	TIP	TIP
240	[103,478]	[68,494]	43.042	TIP	TIP
241	[124,485]	[126,499]	14.828	TIP	TIP
242	[277,491]	[276,503]	13.243	TIP	TIP
243	[328,493]	[323,497]	6.657	TIP	TIP
244	[399,503]	[402,508]	6.828	TIP	TIP